

P02 final-review readiness receipt — 2026-09-23

Status: local preflight passed; final Copilot readiness remains pending. Hosted CI, predecessor acceptance and the late external checkpoint remain open. This report does not accept the full chains feature set or release.

Frozen identity and scope

- Canonical master form SHA256 observed for this reconciliation:
d0be74742b3cc33678f601a92b4ed9c7c345ab4a8a227dedf2f5452cfb6711f4.
- Owner: Codex /root, canonical claim docs/release/audit-claims/715-02/OWNER.md in the main firmware repository.
- Isolated branch: audit/p02-review-readiness-20260923; checkout /private/tmp/kk715-p02-requalification-20260923.
- Adjacent remediation base: 51411c4a7c96137bc80f8b99fa1690c4e385f7fd, release Stack 02 PR #832. The remediation delta is distinct from #832's complete chains diff.
- Firmware/test code head: f5a6fe70c6cf3d46bf4fdb3dfb56ee8df8c4f31. CI coverage head: f5a6fe70c6cf3d46bf4fdb3dfb56ee8df8c4f31. This report is generated from predecessor e19343102de51caa1e7e9c7f2718e2d4a76d7ef2; report-only commits preserve tested code and pins.
- Prior P02 receipt: a6a8b6d73a297ac28181dfcc6e77a1c63ac9b0a6. Its report/PDF supplies original reproductions, historical source identities, the five-contract map and negative-control evidence. This receipt supersedes its open local-blocker dispositions, not its historical test results.
- Finite extension is recorded in P02-readiness-manifest-20260923.md: validated progress, existing unlimited-approval policy, repeatable recovery fixture and isolated predecessor reconciliation.
- Exact merged Stack 01: 23b3c16b1b43e50ff04d8c5e4a77ee37c4a8e393. This supersedes the earlier cec3cac integration after the predecessor owner published review remediation. Stack 00b: 4c56e19e3070772a28b2890a4ec862ecf77d5429. Live PR identities rechecked; neither is accepted by this report. PR #845 and #832 remain non-mergeable in GitHub's reported state.
- Source still declares 7.14.3. Host tests use FW_VERSION=7.14.3; no 7.15 release capability is inferred from the program name.
- Dependency pointers are unchanged from the prior receipt. Host pin remains 49d537ce953b7524599bfc87f56b7a50a51a13a1; no unpublished dependency was imported.

Contract and finding dispositions

ID / contract	Disposition and distinguishing evidence
P02-R01 dispatch boundaries	Fixed in prior P02 unit, retained in merged candidate. Normal and RAW dispatch call before/after hooks; ordinary regressions prove scratch clearing, cross-workflow revocation and top-level confirmation boundaries.
P02-R02 protected Ping and soft Initialize	Fixed and retained. Ping revokes signing before protected waits. Soft Initialize coverage preserves cached PIN while revoking Bitcoin signing and passphrase. Full host Ping/ClearSession and native regressions pass.
P02-R03 unexpected tiny acknowledgements	Fixed and retained. Waiting handlers reject wrong valid ACK types, unwind and wipe copies; one terminal response and immediate Initialize tested over UDP. Original handlers failed negative controls.
P02-R04 short tiny packets	Fixed and retained. UDP/main/debug USB short transfers reject active tiny waits; no-data and independent U2F paths are preserved. Native and raw UDP regressions pass. Physical USB remains a release-only qualification.
Tiny secret and packet lifetime	Inherited wiping explicitly included in review scope: messages.c, confirm_sm.c, pin_sm.c, passphrase_sm.c, usb.c and udp.c. Native stale-secret and packet-storage regressions run in both products.
P02-I01 predecessor conflict	Resolved locally by exact-head merge. Preserve UDP animation and its board regression. Keep explicit validated progress in place of unconditional continuation-response renewal, which would renew on empty recovery/EOS requests. Preserve other chain/reset progress at validated continuation sites. Predecessor acceptance is still external to this claim.

ID / contract	Disposition and distinguishing evidence
P02-I02 workflow progress	Fixed Bitcoin, Ethereum, recovery and EOS missing renewals. Feature polling, invalid Bitcoin ACKs, empty recovery deletion and empty EOS chunks cannot renew. Former guarded regressions now run ordinarily. No KK_FINAL_POLICY_TESTS guards remain in firmware tests.
P02-I02 approval policy	Separate repair restores the existing release-line unlimited ERC-20 refusal, corroborated by a56fb3e88 and the pinned host test. Canonicalize all-zero value before classification; padded zero cannot fall through to generic signing. Finite approvals remain covered by the broad host suite.
P02-I03 warm recovery fixture	Test defect: storage_wipe erases flash but does not reset loaded RAM. Match production WipeDevice's reset/UUID/commit sequence in the fixture. Two complete native suites pass on repeated emulator storage; assertions against committing an empty mnemonic remain.
Predecessor review interactions	Fresh tiny polls clear stale rejection state; the new native regression passes. Both dice consent modes omit raw entropy from DebugLink, with ordinary reset as a working control. Two new host tests pass both products and fail on the older diagnostic emulator at the entropy-presence assertion. Preserve Stack 02 Zcash checked 80-byte nonce, wiping, rk validation and dummy-spend handling instead of replaying the predecessor's older signer call.
F161 / F204	Packet-storage test executes both variants. All local host runs use isolated containers with network none and no published ports.

Local review of the reconciliation

The Stack 01 synthetic TxRequest-only timer test is superseded by a real validated Bitcoin stream test. Parser-level feature polling and a response sent at HOME still cannot defer locking. Explicit renewals preserve Cosmos, Osmosis, Binance, Thorchain, Mayachain, Tendermint and reset continuation behavior after successful initialization/update; invalid branches return first. EOS requires exactly one action and accepted unknown data before renewal. Recovery renews only after actual edits while its ceremony remains armed. SCREENSAVER cannot be undone by the progress hook.

The unlimited-approval check reads the 32-byte allowance only after the standard 68-byte ERC-20 approval classifier succeeds. Native-value storage is bounded by the protobuf field. Dispatch cleanup, failure latches, response arena lifetime and existing dependency pins were retained. The callback API added by Stack 01 remains available, but firmware no longer subscribes to response-only progress inference.

Verification matrix

Gate	Full	Bitcoin-only
Native on reconciled code	428 firmware / 20 board / 18 crypto pass	136 / 20 / 18 pass
Repeated native before merge	425 / 19 / 18 pass twice	Final reconciled suite passes with existing storage
Scoped P02 host transport	15 passed, no skips	15 passed, no skips
Final broad host, including P02 wire tests	580 passed / 257 skipped	351 passed / 486 skipped
Report validator + prompt-unwind + P02	14 passed, no skips	14 passed, no skips
Required screenshot run	68 passed / 19 skipped / 743 deselected	27 passed / 60 skipped / 743 deselected
Screenshot audit	402 PNGs in 68 sequences; pass	177 PNGs in 27 sequences; pass
Production ARM build	Pass	Pass
SRAM stack/heap reserve	18,308 bytes	32,340 bytes
Largest static frame	7,664 bytes; budget pass	7,664 bytes; budget pass
clang-format 20.1.8	All CI-selected C/H paths pass	Same source gate

Builder: kktech/firmware@sha256:7438e53933d47d53157ed6d96d864cb208597e62dce26235ace09d1063427fa2 (Linux amd64). Current host dependency image: sha256:f9b5fb28c71113564bea7f78c1160074155c2da2bc45b162b7dc8892fcc04dba. The older receipt's image is no longer present. Current runs mount the pinned source and audited emulator binaries; they do not test the dependency image's embedded firmware.

A 36-frame contact sheet was visually inspected for readable signing/recovery prompts; this is sampled visual review, not a claim that every chain's screens were manually audited. The machine screenshot audit checked every declared required capture for the configured product/version.

Capabilities and skips

The complete observed test-level skip inventory is `P02-readiness-20260923-skips.json`. Product and version gates are not missing verification disguised as passes. The CI-declared absent capabilities remain: `evm-max-amount-review`, `evm-unknown-token-review`, `entropy-audit-budget`, `osmosis-wire-guards`, `ripple-memo-policy`, `hive-release-review`, `session-trust-lifetime`, `storage-v19-kdf`, `solana-lut-attestation`. These belong to other release units and prevent claiming whole-release acceptance.

Prompt-workflow-unwind was independently executed without its old exclusion and passes both products. CI now removes that obsolete exclusion and explicitly collects the seven P02 transport/predecessor host regressions. The pinned report-validator test `tearDown` clears the capability environment variable; later tests can therefore execute despite the initial exclusion list. This report records observed XML outcomes rather than treating the list as a uniform suite-wide skip count. No tests or dependency pins were weakened to obtain these results.

Storage boot-boundary tests pass in both broad host suites: wallet-preserving reboot, Bitcoin-only band refusal, framed/unframed v16 upgrade and unknown-version wipe. Remaining storage-v19 release controls are separately assigned, not accepted here.

Remaining gates and review preflight

- Exact-head CI run 35830189708 passed every required job for earlier head `ecd5705e42555060ee103d2b0d7089532593c1f9`, with publication intentionally skipped. Its run metadata and artifact IDs/digests are preserved in `predecessor2-prior-ci-run.json` and `predecessor2-prior-ci-artifacts.json`. The new predecessor merge changes code/tests; current-head hosted CI must run after this receipt is sealed. Initial run 35829350176's artifact-checksum false positives remain narrowly corrected.
- Proposed final review preflight is `P02-PR-PREFLIGHT-20260923.md`. The canonical code-bearing script passed for the frozen remediation base/code head; a real final review PR and its upstream-shaped accepted target remain pending.
- Accepted predecessor sequence and internal release acceptance remain pending; another owner's claim is not overridden.
- Freeze a code-bearing upstream-shaped unit at the SOP's late checkpoint, including inherited wiping/Initialize behavior in its review map. Receipt-only PR #850 cannot satisfy this requirement.
- The integration candidate contains coherent transport, progress, policy, fixture and CI commits. Final externally reviewed units must preserve that separation; this receipt does not authorize a broad-product Copilot request.
- Physical-device qualification and release publication are not part of local emulator acceptance.
- No Copilot request has been sent; the active user goal remains in progress.

Evidence and reproduction

Evidence root: `/private/tmp/kk715-p02-audit-evidence`. Artifact hashes are in `P02-readiness-20260923-artifacts.json`. `predecessor2-*xunit.log` records native results; `predecessor2-final-host-*.xml` records final host outcomes; `predecessor2-screenshot-audit-*.log` records required capture coverage; `predecessor2-arm-*sram.log` records memory gates. Host command scripts and stack-usage archives are hashed. Use the exact builder, source pin and product-specific build directories from the prior receipt; the final host runner records the corrected capability mask and actual test arguments.

New unit commit inventories

Counts below use each commit's first-parent diff, including the explicit predecessor merge. Historical transport commit inventories are in the prior receipt. These counts overlap and must not be added to infer the net change.

b786ec9918c78e353964ad10f6133150505f0190

docs(p02): define the final-review readiness batch

Added	Deleted	Path
26	0	docs/release/audit-units/P02-readiness-manifest-20260923.md

2b7054cbb1d400a00e02b16141ab171cfce957db

fix(signing): renew idle deadline on accepted Bitcoin stages

Added	Deleted	Path
19	13	lib/firmware/signing.c
0	2	unittests/firmware/fsm.cpp

4ae9f84ef368a4e04cf5cb2eb0717596156ff754

fix(fsm): renew idle deadline for validated workflow progress

Added	Deleted	Path
2	0	lib/firmware/ethereum.c
6	0	lib/firmware/fsm_msg_eos.h
2	0	lib/firmware/recovery_cipher.c
0	8	unittests/firmware/fsm.cpp

89ea303cc531373dc33b17d3cc9cc270883c5217

test(recovery): reset RAM storage after erasing test flash

Added	Deleted	Path
8	0	unittests/firmware/recovery.cpp

518608729fe7d2f7ca960aaedde411936822c75d

fix(ethereum): restore unlimited approval refusal for padded zero value

Added	Deleted	Path
6	0	docs/release/audit-units/P02-readiness-manifest-20260923.md
21	0	lib/firmware/ethereum.c
0	2	unittests/firmware/fsm.cpp

0646ae6eb30716b9a4be32f50e9f93aa31dd83ae

merge: reconcile Stack 01 validated progress with P02 transport fixes

Added	Deleted	Path
413	0	docs/release/audit-units/715-01-review-20260923.md
-	-	docs/release/audit-units/715-01-review-20260923.pdf
6	0	docs/release/audit-units/P02-readiness-manifest-20260923.md
4	0	include/keepkey/board/messages.h
2	0	lib/board/udp.c
7	0	lib/board/usb.c
2	0	lib/firmware/fsm_msg_binance.h
2	0	lib/firmware/fsm_msg_cosmos.h
2	0	lib/firmware/fsm_msg_mayachain.h
2	0	lib/firmware/fsm_msg_osmosis.h
2	0	lib/firmware/fsm_msg_tendermint.h
2	0	lib/firmware/fsm_msg_thorchain.h
1	0	lib/firmware/reset.c
15	0	unittests/board/board.cpp
48	0	unittests/firmware/fsm.cpp

c0b3250fece02d03845f81d02a404a574cb206bf

ci: enforce P02 transport regressions and prompt unwind capability

Added	Deleted	Path
1	1	.github/workflows/ci.yml
2	1	scripts/emulator/python-keepkey-tests.sh

cee6beacabf406b137a620375a65a41410e94fc2

docs(p02): freeze reconciled local readiness evidence

Added	Deleted	Path
728	0	docs/release/audit-units/P02-readiness-20260923-artifacts.json
3033	0	docs/release/audit-units/P02-readiness-20260923-skips.json
198	0	docs/release/audit-units/P02-readiness-20260923.md
-	-	docs/release/audit-units/P02-readiness-20260923.pdf

fa9bb4f5915edb7f44fc1e400e816ae77dc997bc

docs(p02): include readiness artifacts in final adjacent inventory

Added	Deleted	Path
6	2	docs/release/audit-units/P02-readiness-20260923.md
-	-	docs/release/audit-units/P02-readiness-20260923.pdf

139c546796fbf63d58e48e23b157a5b03b788b26

docs(p02): seal stable readiness inventory

Added	Deleted	Path
3	3	docs/release/audit-units/P02-readiness-20260923.md
-	-	docs/release/audit-units/P02-readiness-20260923.pdf

cfaa4d282f429ff26a4d45c680758f35ad80fd85

ci: recognize two recorded P02 ELF hashes as evidence checksums

Added	Deleted	Path
8	0	.gitleaks.toml

38ae657b097cb560db24179fbdef3cdfa390a75

docs(p02): record CI checksum false-positive disposition

Added	Deleted	Path
2	0	docs/release/audit-units/P02-readiness-20260923-artifacts.json
41	3	docs/release/audit-units/P02-readiness-20260923.md
-	-	docs/release/audit-units/P02-readiness-20260923.pdf

095abe7fd221a4434156afad85394921e1b31087

docs(p02): refresh corrected CI candidate inventory

Added	Deleted	Path
4	4	docs/release/audit-units/P02-readiness-20260923.md
-	-	docs/release/audit-units/P02-readiness-20260923.pdf

ecd5705e42555060ee103d2b0d7089532593c1f9

docs(p02): seal corrected CI candidate receipt

Added	Deleted	Path
2	2	docs/release/audit-units/P02-readiness-20260923.md
-	-	docs/release/audit-units/P02-readiness-20260923.pdf

f5a6fe70c6cf3d46bf4fdb3dfef56ee8df8c4f31

merge: retain Stack 01 review fixes across P02 transport and dice waits

Added	Deleted	Path
33	1	docs/release/audit-units/715-01-review-20260923.md
-	-	docs/release/audit-units/715-01-review-20260923.pdf
6	0	docs/release/audit-units/P02-readiness-manifest-20260923.md
1	1	docs/security/7.14.3-bitcoin-only-dice-audit-sop.md
1	0	lib/board/messages.c
1	1	lib/firmware/fsm_msg_debug.h
6	1	lib/firmware/reset.c
48	0	unittests/firmware/usb_rx.cpp
26	0	unittests/host/test_p02_transport.py

Complete adjacent inventory

Generation predecessor: e19343102de51caa1e7e9c7f2718e2d4a76d7ef2. Reproduce with `git diff --numstat 51411c4a7c96137bc80f8b99fa1690c4e385f7fd..e19343102de51caa1e7e9c7f2718e2d4a76d7ef2`. Final report-only regeneration must preserve these path/count pairs, including report artifacts.

Added	Deleted	Path
1	1	.github/workflows/ci.yml
8	0	.gitleaks.toml
445	0	docs/release/audit-units/715-01-review-20260923.md
-	-	docs/release/audit-units/715-01-review-20260923.pdf
42	0	docs/release/audit-units/P02-PR-PREFLIGHT-20260923.md
780	0	docs/release/audit-units/P02-readiness-20260923-artifacts.json
3033	0	docs/release/audit-units/P02-readiness-20260923-skips.json
290	0	docs/release/audit-units/P02-readiness-20260923.md
-	-	docs/release/audit-units/P02-readiness-20260923.pdf
44	0	docs/release/audit-units/P02-readiness-manifest-20260923.md
36	0	docs/release/audit-units/P02-requalification-20260923-artifacts.json
246	0	docs/release/audit-units/P02-requalification-20260923.md
-	-	docs/release/audit-units/P02-requalification-20260923.pdf
1	1	docs/security/7.14.3-bitcoin-only-dice-audit-sop.md
8	0	include/keepkey/board/messages.h
5	1	lib/board/confirm_sm.c

```

36 1 lib/board/messages.c
2 0 lib/board/udp.c
15 2 lib/board/usb.c
23 0 lib/firmware/ethereum.c
2 0 lib/firmware/fsm_msg_binance.h
10 0 lib/firmware/fsm_msg_common.h
2 0 lib/firmware/fsm_msg_cosmos.h
1 1 lib/firmware/fsm_msg_debug.h
6 0 lib/firmware/fsm_msg_eos.h
2 0 lib/firmware/fsm_msg_mayachain.h
2 0 lib/firmware/fsm_msg_osmosis.h
2 0 lib/firmware/fsm_msg_tendermint.h
2 0 lib/firmware/fsm_msg_thorchain.h
8 0 lib/firmware/passphrase_sm.c
3 0 lib/firmware/pin_sm.c
2 0 lib/firmware/recovery_cipher.c
7 1 lib/firmware/reset.c
19 13 lib/firmware/signing.c
2 1 scripts/emulator/python-keepkey-tests.sh
15 0 unittests/board/board.cpp
48 20 unittests/firmware/fsm.cpp
8 0 unittests/firmware/recovery.cpp
21 0 unittests/firmware/storage_passphrase.cpp
158 1 unittests/firmware/usb_rx.cpp
74 0 unittests/host/test_p02_transport.py

```